Nathan R. Tallent

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Research Summary

Dr. Nathan Tallent is a distinguished computer scientist within the Future Computing Technologies Group at Pacific Northwest National Laboratory. His research is motivated by emerging challenges in distributed systems, machine learning, scientific workflows, and data management. He leads activities in continuum computing and the Performance Lab for EXtreme Computing and daTa where his contributions have spanned the challenges of performance measurement, modeling, bottleneck diagnosis, and optimization; and includes special attention to bottlenecks in networks, storage, and memory. He has made notable contributions to performance tools, both for performance modeling and for parallel performance analysis. He has more than 80 peer-reviewed publications, serves on several reviewing committees, and received a DOE Early Career award.

Dr. Tallent is leading ASCR projects on performance prediction and diagnosis for extreme-scale scientific workflows. His work partners with ASCR's AMAIS ("Advanced Memory to Support Artificial Intelligence for Science") and ASCR's ENCODE ("End-to-end co-design for performance, energy efficiency, and security in AI-enabled computational science") project in characterizing workloads, generating performance models, analyzing data movement, diagnosing performance bottlenecks, and evaluating new technology. He has led development of several research software prototypes for distributed scientific workflows, distributed AI services, and performance analysis and prediction. In particular, he leads development of DataFlowDrs, a measurement and analysis toolsets for distributed scientific workflows; and the MemGaze the Palm performance analysis and modeling tools. He led development of the SEAK and PERFECT benchmark suites, designed for evaluating performance-power-architecture tradeoffs. He was an original developer of HPCToolkit, a widely used suite of performance tools.

Education

Ph.D., Computer Science, *Rice University*, Houston, TX — May 2010

M.S., Computer Science, *Rice University*, Houston, TX — May 2007

M.Div., *Westminster Theological Seminary*, Philadelphia, PA — May 2002

B.A., Computer Science, *Rice University*, Houston, TX — May 1998

Selected Work

◇ US DOE Early Career (2021)

◇ ***IPDPS*** 26a, 26b, 26c, 26d, 26e; 25a, 25b; 24, 23, 17, 16, 16 • ***SC*** 23, 21, 17, 15, 10, 09 • ***ICS*** 26, 25, 14, 11
CLUSTER 24a, 24b, 18, 22 • ***IISWC*** 20, 18 • ***ISPASS*** 20 • ***PPoPP*** 15, 10, 09
AAAI 26 • ***ICDM*** 25 • ***BigData*** 20, 19 • ***PLDI*** 09
JPDC 23 • ***TPDS*** 21, 20 • ***C&C*** 10 • ***IEEE Computer*** 09

◇ Best paper nominees: ICS 25, SSDBM 25, IISWC 18, SC 15, PLDI 09.

◇ ACM/IEEE-CS George Michael Memorial HPC Fellowship (2009)

Professional Experience

◇ Chief Computer Scientist, Pacific Northwest National Laboratory, Jan. 2022–present.

◇ Senior Computer Scientist, Pacific Northwest National Laboratory, Oct. 2011–2021.

◇ Research Scientist, Dept. of Computer Science, Rice University, Apr. 2010–Oct. 2011.

◇ Performance Tools Consultant (Samara Technology Group, SiCortex), Jan. 2007–Mar. 2011.

Professional Leadership

- ◇ PI, DOE ASCR (Early Career) "Orchestration for Distributed and Data-Intensive Scientific Exploration," 2021-2026.
- ◇ Co-PI, AT SCALE (LDRD) "Data-Intensive Scientific Exploration", 2024-25.
- ◇ Chief Scientist, PNNL Agile investment "Cloud, HPC, and Edge for Science and Security" (CHESS), 2022-24.
- ◇ Co-PI, DMC (LDRD) "Fixing Amdahl's Law within the Limits of Accelerated Systems" (Fallacy), 2019-22.
- ◇ PI, DOE ASCR "Integrated End-to-end Performance Prediction and Diagnosis for Extreme Scientific Workflows" (IPPD/2), 2017-2020.
- ◇ Co-PI, DOE ASCR "Performance Insight for Programmers and Exascale Runtimes" (PIPER), 2012-16.
- ◇ Chair of Performance track, ICPP 2024 (53rd International Conference on Parallel Processing
- ◇ Chair of Performance Modeling, Evaluation, and Analysis Track, ISC 2022 (Intl. Supercomputing Conf.) 2022.
- ◇ Co-Organizer, Workshop on Principles of Memory Hierarchy Optimization, in conjunction with PPoPP (Principles and Practice of Parallel Programming) 2022.
- ◇ Workshop/Tutorial Chair, IEEE International Symposium on Workload Characterization 2022.
- ◇ Co-chair, Performance Track, SC (Intl. Conf. for High Performance Computing, Networking, Storage and Analysis), 2019.
- ◇ Co-Organizer, Workshop on Suite of Embedded Applications and Kernels, in conjunction with DAC (Design Automation Conf.) 2015.
- ◇ Co-Organizer, The Second Intl. Workshop on High-performance Infrastructure for Scalable Tools, in conjunction with ICS (Intl. Conf. on Supercomputing) 2012.
- ◇ Co-Organizer, The First Intl. Workshop on High-performance Infrastructure for Scalable Tools, in conjunction with ICS (Intl. Conf. on Supercomputing) 2011.

Professional Service

- ◇ Western Washington University, Dept. of Computer Science Advisory Board
- ◇ Program Committee, 32nd ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming, 2027 (PPoPP 2027)
- ◇ Program Committee, IEEE/ACM Intl. Conf. for High Performance Computing, Networking, Storage and Analysis, 2026 (SC25)
- ◇ Program Committee, ACM 35rd International Symposium on High-Performance Parallel and Distributed Computing (HPDC 2026)
- ◇ 1st Workshop On Data Reduction And Energy-Aware Data Movement (Dream) SCA 2026
- ◇ Program Committee, 40th IEEE International Parallel and Distributed Processing Symposium (IPDPS 2026)
- ◇ Program Committee, 53rd International Conference on Parallel Processing (ICPP 2025)
- ◇ Program Committee, IEEE/ACM Intl. Conf. for High Performance Computing, Networking, Storage and Analysis, 2025 (SC25)
- ◇ Program Committee, ACM 34rd International Symposium on High-Performance Parallel and Distributed Computing (HPDC 2025)
- ◇ Program Committee, 30th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP 2025)
- ◇ ACM SIGHPC Doctoral Dissertation Award Committee
- ◇ Program Committee, ACM 33rd International Symposium on High-Performance Parallel and Distributed Computing (HPDC 2024)
- ◇ Program Committee, IEEE/ACM Intl. Conf. for High Performance Computing, Networking, Storage and Analysis,

2024 (SC24)

- ◇ Program Committee, 53rd International Conference on Parallel Processing (ICPP 2024)
- ◇ Program Committee, 30th International European Conference on Parallel and Distributed Computing (Euro-PAR 2024)
- ◇ Program Committee, 38th IEEE International Parallel and Distributed Processing Symposium (IPDPS'24)
- ◇ Program Committee, 29th ACM SIGPLAN Symposium on Principles and Practice of Parallel Programming (PPoPP 2024)
- ◇ Program Committee, 10th IEEE/ACM International Conference on Big Data Computing, Applications and Technologies (BDCAT2023)
- ◇ Program Committee, 2023 Workshop on Programming and Performance Visualization Tools (ProTools)
- ◇ Program Committee, IEEE/ACM Intl. Conf. for High Performance Computing, Networking, Storage and Analysis, 2023 (SC23)
- ◇ Program Committee, IEEE 37th Intl. Parallel and Distributed Processing Symp., 2023
- ◇ Program Committee, Intl. Supercomputing Conf., 2023
- ◇ Program Committee, 37th ACM International Conference on Supercomputing, 2023 (ICS).
- ◇ Program Committee, IEEE 37th Intl. Parallel and Distributed Processing Symp., 2023 (IPDPS).
- ◇ Program Committee, IEEE/ACM Intl. Conf. for High Performance Computing, Networking, Storage and Analysis, 2022.
- ◇ Program Committee, 2022 Workshop on Programming and Performance Visualization Tools (ProTools)
- ◇ Program Committee, 51th International Conference on Parallel Processing, 2022.
- ◇ Program Committee, IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing, 2022
- ◇ Program Committee, IEEE 36th Intl. Parallel and Distributed Processing Symp., 2022
- ◇ Program Committee, Supercomputing Frontiers Asia 2022
- ◇ 2021 Program Committee, Workshop on Programming and Performance Visualization Tools (ProTools)
- ◇ Program Committee, IEEE/ACM Intl. Conf. for High Performance Computing, Networking, Storage and Analysis, 2021.
- ◇ Program Committee, 50th International Conference on Parallel Processing, 2021
- ◇ Program Committee, IEEE 35th Intl. Parallel and Distributed Processing Symp., 2021
- ◇ 2020 Program Committee, Workshop on Programming and Performance Visualization Tools (ProTools)
- ◇ Program Committee, Intl. Conf. on Massive Storage Systems and Technology, 2020.
- ◇ Program Committee, Supercomputing Asia, 2020.
- ◇ Program Committee, IEEE/ACM Intl. Symp. on Cluster, Cloud and Grid Computing, 2020.
- ◇ Program Committee, IEEE Cluster Conf., 2019.
- ◇ Program Committee, Intl. Conf. on Parallel Processing, 2019.
- ◇ Extended Program Committee, ACM/SIGARCH Intl. Conf. on Supercomputing, 2019.
- ◇ Program committee, IEEE/ACM Intl. Symp. on Cluster, Cloud and Grid Computing, 2019.
- ◇ Program Committee, IEEE/ACM Intl. Conf. for High Performance Computing, Networking, Storage and Analysis, 2018.
- ◇ Program Committee, ACM/SIGARCH Intl. Conf. on Supercomputing, 2018.
- ◇ Program Committee, IEEE/ACM Intl. Symp. on Cluster, Cloud and Grid Computing, 2018.

- ◇ Program Committee, IEEE Cluster Conf., 2018.
- ◇ Program Committee, Intl. Conf. on Parallel Processing, 2018.
- ◇ Program committee, Supercomputing Asia, 2018.
- ◇ Program Committee, IEEE Cluster Conf., 2017.
- ◇ Program Committee, IEEE/ACM Intl. Conf. for High Performance Computing, Networking, Storage and Analysis, Nov. 2016.
- ◇ Program Committee, IEEE/ACM Intl. Symp. on Cluster, Cloud and Grid Computing, 2016.
- ◇ Program Committee, IEEE 30th Intl. Parallel and Distributed Processing Symp., 2016.
- ◇ Program Committee, ACM/SIGARCH 29th Intl. Conf. on Supercomputing, 2015.
- ◇ Program Committee, 15th IEEE/ACM Intl. Symp. on Cluster, Cloud and Grid Computing, 2015.
- ◇ Program Committee: Fifth Intl. Workshop on Parallel Software Tools and Tool Infrastructures (part of ICPP 2014), 2014.
- ◇ Program Committee, PROPER 2014: 7th Workshop on Productivity and Performance (part of EuroPar 2014), 2014.
- ◇ Extended Review Committee, ICS 2014: The ACM/SIGARCH 28th Intl. Conf. on Supercomputing, 2014.
- ◇ Program Committee, DAC Workshop on Suite of Embedded Applications and Kernels (part of 51st Design Automation Conf.), 2014.
- ◇ Program Committee, HIPS 2014: 19th Intl. Workshop on High-Level Parallel Programming Models and Supportive Environments (part of IPDPS 2014), 2014.
- ◇ Birds of a Feather Committee Member, SC 2013: The IEEE/Computer and ACM/SIGARCH 2013 Intl. Conf. for High Performance Computing, Networking, Storage and Analysis, 2013.
- ◇ Program Committee, PSTI 2013: Fourth Intl. Workshop on Parallel Software Tools and Tool Infrastructures (part of ICPP 2013), Lyon, France, October, 2013.
- ◇ Program Committee, PROPER 2013: 6th Workshop on Productivity and Performance (part of EuroPar 2013), 2013.
- ◇ Program Committee, ASPLOS 2013: The ACM 18th Intl. Conf. on Architectural Support for Programming Languages and Operating Systems, 2013.
- ◇ Program Committee, PPoPP 2013: The ACM/SIGPLAN 18th Symp. on Principles and Practice of Parallel Programming, 2013.
- ◇ Program Committee, PSTI 2012: Third Intl. Workshop on Parallel Software Tools and Tool Infrastructures (part of ICPP 2012), 2012.
- ◇ External Review Committee, PPoPP 2012: The ACM/SIGPLAN 17th Symp. on Principles and Practice of Parallel Programming, 2012.
- ◇ Program Committee, SC 2011: The IEEE/Computer and ACM/SIGARCH 2011 Intl. Conf. for High Performance Computing, Networking, Storage and Analysis, 2011.
- ◇ Program Committee, PSTI 2011: Second Intl. Workshop on Parallel Software Tools and Tool Infrastructures (part of ICPP 2011), 2011.
- ◇ Program Committee, ICS 2011: The ACM/SIGARCH 25th Intl. Conf. on Supercomputing, 2011.
- ◇ Program Committee, CGO 2011: 2011 Intl. Symp. on Code Generation and Optimization, 2011.
- ◇ Reviewer for *ACM Transactions on Architecture and Code Optimization*, *ACM Transactions on Parallel Computing*, *Concurrency and Computation: Practice and Experience*, *IEE Computer*, *IEEE Transactions on Computers*, *IEEE Transactions on Parallel and Distributed Systems*, *Journal of Systems and Software*, *Journal of Parallel and Distributed Computing*, *Parallel Computing: Systems & Applications*, *Software: Practice and Experience*.

Publications: Distributed AI Systems and Data Analytics

- ◇ Sarkar, Aishwarya, Ghosh, Sayan, **Tallent, Nathan R.**, Chadha, Aman, Roosta, Tanya G., and Jannesari, Ali. **June 2026**. "Rudder: Steering Prefetching in Distributed GNN Training using LLM Agents". In: *Proc. of the 40th ACM Intl. Conf. on Supercomputing*. ICS '26. New York, NY, USA: Association for Computing Machinery. DOI: 10.1145/3797905.3800542.
- ◇ Li, Boqiang, Guo, Luanzheng, She, Buxin, **Tallent, Nathan R.**, Adetola, Veronica, and Ge, Rong. **May 2026**. "PowerMorph: Shaping LLM Training for Data Center Demand Response". In: *Proc. of the 40th IEEE Intl. Parallel and Distributed Processing Symp.* IPDPS '26. IEEE Computer Society.
- ◇ Zhang, Boyuan, **Tallent, Nathan R.** et al. **May 2026**. "Accelerating AI Compression through Lightweight Lossless Encoding and Pipelined Workflows". In: *Proc. of the 40th IEEE Intl. Parallel and Distributed Processing Symp.* IPDPS '26. IEEE Computer Society.
- ◇ Gong, Chengyu, Shen, Gefei, Guo, Luanzheng, **Tallent, Nathan R.**, and Zhao, Dongfang. **Jan. 2026**. "Order-Preserving Dimension Reduction for Multimodal Semantic Embedding". In: *Proc. of the AAAI Conference on Artificial Intelligence.* AAAI 40.
- ◇ Mehboob, Talha, Guo, Luanzheng, **Tallent, Nathan R.**, Zink, Michael, and Irwin, David. **Nov. 2025**. "PowerTrip: Exploiting Federated Heterogeneous Datacenter Power for Distributed ML Training". In: *Proc. of the 2025 ACM Symposium on Cloud Computing.* SoCC '25. New York, NY, USA: Association for Computing Machinery, pp. 762–775. DOI: 10.1145/3772052.3772228.
- ◇ Fu, Jiuzhou, Guo, Luanzheng, **Tallent, Nathan R.**, and Zhao, Dongfang. **Nov. 2025**. "ProHD: Projection-Based Hausdorff Distance Approximation". In: *Proc. of the 25th IEEE Intl. Conf. on Data Mining*. ICDM '25.
- ◇ Sun, Minqiu, Huang, Xin, Guo, Luanzheng, **Tallent, Nathan R.**, Sato, Kento, and Dai, Dong. **Nov. 2025**. "LLMTailor: A Layer-wise Tailoring Tool for Efficient Checkpointing of Large Language Models". In: *Proc. of the SC '25 Workshops of the Intl. Conf. for High Performance Computing, Networking, Storage and Analysis (10th Intl Parallel Data Systems Workshop).* SC Workshops '25. New York, NY, USA: Association for Computing Machinery, pp. 1366–1374. DOI: 10.1145/3731599.3767515.
- ◇ Faykus, Max H., Guo, Luanzheng, Ashraf, Rizwan A., **Tallent, Nathan R.**, and Calhoun, Jon C. **June 2025**. "Exploration of LLM Lossless Compression on Scientific Data". In: *IEEE Intl. Workshop on HPC for AI Foundation Models and LLMs for Science (co-located with 39th IEEE Intl. Parallel and Distributed Processing Symp.)* IPDPS Workshops '25. IEEE Computer Society, pp. 986–990. DOI: 10.1109/IPDPSW66978.2025.00153.
- ◇ Abebe, Waqwoya, Strube, Jan, Guo, Luanzheng, **Tallent, Nathan R.**, Bel, Oceane, Spurgeon, Steven, Doty, Christina, and Jannesari, Ali. **Jan. 2025**. "SAM-I-Am: Semantic boosting for zero-shot atomic-scale electron micrograph segmentation". In: *Computational Materials Science.* Comput. Mater. Sci. 246, p. 113400. DOI: 10.1016/j.commatsci.2024.113400.
- ◇ Sarkar, Aishwarya, Ghosh, Sayan, **Tallent, Nathan R.**, and Jannesari, Ali. **Sept. 2024**. "MassiveGNN: Efficient Training via Prefetching for Massively Connected Distributed Graphs". In: *Proc. of the 2024 IEEE Conf. on Cluster Computing*. CLUSTER '24. IEEE, pp. 62–73. DOI: 10.1109/CLUSTER59578.2024.00013.
- ◇ Chen, Jou-An, Sung, Hsin-Hsuan, Shen, Xipeng, **Tallent, Nathan**, Barker, Kevin, and Li, Ang. **2023e**. "Accelerating matrix-centric graph processing on GPUs through bit-level optimizations". In: *Journal of Parallel and Distributed Computing*. JPDC 177, pp. 53–67. DOI: 10.1016/j.jpdc.2023.02.013.
- ◇ Chen, Jou-An, Sung, Hsin-Hsuan, **Tallent, Nathan R.**, Barker, Kevin, Shen, Xipeng, and Li, Ang. **May 2022**. "Bit-GraphBLAS: Bit-Level Optimizations of Matrix-Centric Graph Processing on GPU". in: *Proc. of the 36st IEEE Intl. Parallel and Distributed Processing Symp.* IPDPS '22. IEEE. DOI: 10.1109/IPDPS53621.2022.00056.
- ◇ Li, Ang, Song, S., Chen, J., Li, J., Liu, X., **Tallent, Nathan**, and Barker, Kevin J. **Jan. 2020**. "Evaluating Modern GPU Interconnect: PCIe, NVLink, NV-SLI, NVSwitch and GPUDirect". In: *IEEE Transactions on Parallel and Distributed Systems.* TPDS 31.1, pp. 94–110. DOI: 10.1109/TPDS.2019.2928289.
- ◇ Li, Ang, Song, Shuaiwen Leon, Liu, Xu, **Tallent, Nathan**, and Barker, Kevin. **Sept. 2018**. "Tartan: Evaluating Modern GPU Interconnect via a Multi-GPU Benchmark Suite". In: *Proc. of the 2018 IEEE Intl. Symp. on Workload Characterization.* IISWC '18. Best Paper Finalist, pp. 191–202. DOI: 10.1109/IISWC.2018.8573483.

◇ **Tallent, Nathan R.**, Gawande, Nitin A., Siegel, Charles, Vishnu, Abhinav, and Hoisie, Adolfy. **Dec. 2017**. "Evaluating On-Node GPU Interconnects for Deep Learning Workloads". In: *High Performance Computing Systems. Performance Modeling, Benchmarking, and Simulation.* Ed. by Stephen Jarvis, Steven Wright, and Simon Hammond. SC PMBS '17. Springer International Publishing, pp. 3–21. DOI: 10.1007/978-3-319-72971-8_1.

Publications: Distributed Scientific Workflows

◇ Rashid, Md Hasanur, Firoz, Jesun, **Tallent, Nathan R.**, Guo, Luanzheng, Tang, Meng, and Dai, Dong. **May 2026**. "QoSFlow: Ensuring Service Quality of Distributed Workflows Using Interpretable Sensitivity Models". In: *Proc. of the 40th IEEE Intl. Parallel and Distributed Processing Symp.* IPDPS '26. IEEE Computer Society.

◇ Tang, Meng, Guo, Luanzheng, Kougkas, Anthony, Sun, Xian-He, and **Tallent, Nathan R. May 2026**. "Characterization and Implications of Dataflow in HPC Workflows". In: *Proc. of the 40th IEEE Intl. Parallel and Distributed Processing Symp.* IPDPS '26. IEEE Computer Society.

◇ Rashid, Md Hasanur, **Tallent, Nathan R.**, Bao, Forrest Sheng, and Dai, Dong. **May 2026**. "CARAT: Client-Side Adaptive RPC and Cache Co-Tuning for Parallel File Systems". In: *Proc. of the 40th IEEE Intl. Parallel and Distributed Processing Symp.* IPDPS '26. IEEE Computer Society.

◇ Tang, Meng, Zhu, Zhaobin, Guo, Luanzheng, Bandy, James G., Carlson, Tim, Neuwirth, Sarah, Kougkas, Anthony, Sun, Xian-He, and **Tallent, Nathan R. Nov. 2025**. "Quantifying AWS S3 I/O Performance Boundaries Using the Roofline Model". In: *Proc. of the SC '25 Workshops of the Intl. Conf. for High Performance Computing, Networking, Storage and Analysis (10th Intl Parallel Data Systems Workshop).* SC Workshops '25. New York, NY, USA: Association for Computing Machinery, pp. 1415–1423. DOI: 10.1145/3731599.3767513.

◇ Zhang, Boyuan, Fang, Bo, Ye, Fanjiang, Guo, Luanzheng, Song, Fengguang, **Tallent, Nathan R.**, and Tao, Dingwen. **June 2025**. "BMQSim: Overcoming Memory Constraints in Quantum Circuit Simulation with a High-Fidelity Compression Framework". In: *Proc. of the 39th ACM Intl. Conf. on Supercomputing.* ICS '25. Best Paper Runner Up. New York, NY, USA: Association for Computing Machinery, pp. 689–704. DOI: 10.1145/3721145.3725747.

◇ Firoz, Jesun, Lee, Hyungro, Guo, Luanzheng, Tang, Meng, **Tallent, Nathan R.**, and Peng, Zhen. **June 2025**. "FastFlow: Rapid Workflow Response by Prioritizing Critical Data Flows and their Interactions". In: *Proc. of the 37th Intl. Conf. on Scalable Scientific Data Management.* SSDBM '25. Best Paper Nominee. ACM, pp. 1–12. DOI: 10.1145/3733723.3733735.

◇ Lee, Hyungro, Firoz, Jesun, **Tallent, Nathan R.**, Guo, Luanzheng, and Halappanavar, Mahantesh. **June 2025**. "FlowForecaster: Automatically Inferring Detailed & Interpretable Workflow Scaling Models for Forecasts". In: *Proc. of the 39th IEEE Intl. Parallel and Distributed Processing Symp.* IPDPS '25. IEEE Computer Society, pp. 420–432. DOI: 10.1109/IPDPS64566.2025.00045.

◇ Newaz, Nahid, Ghosh, Sayan, **Tallent, Nathan R.**, and Qu, Guangzhi. **June 2025**. "Locality Aware Process Remapping for Distributed-Memory Graph Workloads". In: *Proc. of the 39th IEEE Intl. Parallel and Distributed Processing Symp.* IPDPS '25. IEEE Computer Society, pp. 447–459. DOI: 10.1109/IPDPS64566.2025.00047.

◇ Guo, Luanzheng, Tang, Meng, Lee, Hyungro, Firoz, Jesun, and **Tallent, Nathan R. Dec. 2024**. "Improving I/O-aware Workflow Scheduling via Data Flow Characterization and Trade-off Analysis". In: *Seventh IEEE Intl. Workshop on Benchmarking, Performance Tuning and Optimization for Big Data Applications (Proc. of the IEEE Intl. Conf. on Big Data).* Big Data Workshops '24. IEEE Computer Society, pp. 3674–3681. DOI: 10.1109/BigData62323.2024.10825855.

◇ Egersdoerfer, Chris, Rashid, Md. Hasanur, Dai, Dong, Fang, Bo, and **Tallent, Nathan R. Nov. 2024**. "Understanding and Predicting Cross-Application I/O Interference in HPC Storage Systems". In: *Proc. of the Workshops of the Intl. Conf. for High Performance Computing, Networking, Storage and Analysis (9th Intl. Parallel Data Systems Workshop).* SC Workshops '24. DOI: 10.1109/SCW63240.2024.00174.

◇ Tang, Meng, Cernuda, Jaime, Ye, Jie, Guo, Luanzheng, **Tallent, Nathan R.**, Kougkas, Anthony, and Sun, Xian-He. **Sept. 2024**. "DaYu: Optimizing Distributed Scientific Workflows by Decoding Dataflow Semantics and Dynamics". In: *Proc. of the 2024 IEEE Conf. on Cluster Computing.* CLUSTER '24. IEEE, pp. 357–369. DOI: 10.1109/CLUSTER59578.2024.00038.

- ◇ Lee, Hyungro, Guo, Luanzheng, Tang, Meng, Firoz, Jesun, **Tallent, Nathan**, Kougkas, Anthony, and Sun, Xian-He. **Nov. 2023**. "Data Flow Lifecycles for Optimizing Workflow Coordination". In: *Proc. of the Intl. Conf. for High Performance Computing, Networking, Storage and Analysis (SuperComputing).* SC '23. New York, NY, USA: Association for Computing Machinery. DOI: 10.1145/3581784.3607104.
- ◇ Acer, Seher, **Tallent, Nathan R.** et al. **Sept. 2021**. "EXAGRAPH: Graph and Combinatorial Methods for Enabling Exascale Applications". In: *International Journal of High Performance Computing Applications.* Int. J. High Perform. Comput. Appl. P. 10943420211029299. DOI: 10.1177/10943420211029299.
- ◇ Bel, Oceane, Mukhopadhyay, Sinjoni, **Tallent, Nathan R.**, Nawab, Faisal, and Long, Darrell. **Dec. 2021**. "WinnowML: Stable feature selection for maximizing prediction accuracy of time-based system modeling". In: *Proc. of the 2021 IEEE Intl. Conf. on Big Data (Fifth IEEE Intl. Workshop on Benchmarking, Performance Tuning and Optimization for Big Data Applications).* Big Data Workshops '21, pp. 3031–3041. DOI: 10.1109/BigData52589.2021.9671602.
- ◇ Bel, Oceane, Pata, Joosep, Vlimant, Jean-Roch, **Tallent, Nathan**, Balcas, Justas, and Spiropulu, Maria. **May 2021**. "Diolkos: Improving ethernet throughput through dynamic port selection". In: *18th ACM International Conference on Computing Frontiers.* Computing Frontiers '21. New York, NY, USA: ACM. DOI: 10.1145/3457388.3458659.
- ◇ Friese, Ryan D., Mutlu, Burcu O., **Tallent, Nathan R.**, Suetterlein, Joshua, and Strube, Jan. **Dec. 2020**. "Effectively Using Remote I/O For Work Composition in Distributed Workflows". In: *Proc. of the 2020 IEEE Intl. Conf. on Big Data.* BigData '20. IEEE Computer Society, pp. 426–433. DOI: 10.1109/BigData50022.2020.9378352.
- ◇ Bel, Oceane, Chang, Kenneth, **Tallent, Nathan R.**, Duellmann, Dirk, Miller, Ethan L., Nawab, Faisal, and Long, Darrell D. E. **Oct. 2020**. "Geomancy: Automated Performance Enhancement through Data Layout Optimization". In: *36th Intl. Conf. on Massive Storage Systems and Technology.* MSST '20.
- ◇ Suetterlein, Joshua, Friese, Ryan D., **Tallent, Nathan R.**, and Schram, Malachi. **Dec. 2019**. "TAZeR: Hiding the Cost of Remote I/O in Distributed Scientific Workflows". In: *Proc. of the 2019 IEEE Intl. Conf. on Big Data.* BigData '19. IEEE Computer Society, pp. 383–394. DOI: 10.1109/BigData47090.2019.9006418.
- ◇ Schram, Malachi, **Tallent, Nathan**, Friese, Ryan, and Singh, Alok. **Sept. 2019**. "Application of Deep Learning on Integrating Prediction, Provenance, and Optimization". In: *EPJ Web of Conferences.* Ed. by Peter Hristov, Latchezar Betev, and Maarten Litmaath. Vol. 214. EPJ Web Conf. DOI: 10.1051/epjconf/201921406007.
- ◇ Bhuiyan, Tanveer Hossain, Halappanavar, Mahantesh, Friese, Ryan D., Medal, Hugh, de la Torre, Luis, Sathanur, Arun, and **Tallent, Nathan R. 2019e**. "Stochastic Programming Approach for Resource Selection Under Demand Uncertainty". In: *Job Scheduling Strategies for Parallel Processing.* Ed. by Dalibor Klusáček, Walfredo Cirne, and Narayan Desai. JSSPP '19. Cham: Springer International Publishing, pp. 107–126. DOI: 10.1007/978-3-030-10632-4_6.
- ◇ Singh, Alok, Altintas, Ilkay, Schram, Malachi, and **Tallent, Nathan**. **Dec. 2018**. "Deep Learning for Enhancing Fault Tolerant Capabilities of Scientific Workflows". In: *Second IEEE Intl. Workshop on Benchmarking, Performance Tuning and Optimization for Big Data Applications (Proc. of the IEEE Intl. Conf. on Big Data).* Big Data Workshops '18, pp. 3905–3914. DOI: 10.1109/BigData.2018.8622509.
- ◇ Friese, Ryan D., **Tallent, Nathan R.**, Schram, Malachi, Halappanavar, Mahantesh, and Barker, Kevin J. **Sept. 2018**. "Optimizing Distributed Data-Intensive Workflows". In: *Proc. of the 2018 IEEE Conf. on Cluster Computing.* CLUSTER '18. IEEE, pp. 279–289. DOI: 10.1109/CLUSTER.2018.00045.
- ◇ Schram, Malachi, Bansal, Vikas, Friese, Ryan D., **Tallent, Nathan R.**, Yin, Jian, Barker, Kevin J., Stephan, Eric, Halappanavar, Mahantesh, and Kerbyson, Darren J. **Oct. 2017**. "Integrating Prediction, Provenance, and Optimization into High Energy Workflows". In: *Journal of Physics: Conference Series* 898.6, p. 062052. DOI: 10.1088/1742-6596/898/6/062052.
- ◇ Halappanavar, Mahantesh, Schram, Malachi, de La Torre, Luis, Barker, Kevin, **Tallent, Nathan R.**, and Kerbyson, Darren. **Nov. 2015**. "Towards Efficient Scheduling of Data Intensive High Energy Physics Workflows". In: *WORKS '15: Workshop on Workflows in Support of Large-Scale Science, held in conjunction with SuperComputing 15.* SC Workshops '15. DOI: 10.1145/2822332.2822335.

Publications: Hardware/Software Co-design

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Mentoring, Advising

- Mentoring for more than 30 Post Doctoral Researchers and Interns
- Ph.D. Committee, Hasanur Rashid, University of Delaware

- Ph.D. Committee, Yasodha Suriyakumar, Portland State University
- Ph.D. Committee, Oceane Bel, University of California, Santa Cruz